



Service Tool Instruction

Service Tool Instruction
Number: 3377791-26

Revision Level: 26

Released Date: 16-Dec-
2019

Description

ECM Bench Calibration Base Harness

Purpose

This service tool bulletin provides information for the engine control module (ECM) bench calibration base harness, Part Number 3163151. The base harness is used with the appropriate bench calibration adapter harness (see Table 3) to calibrate an ECM without installing it on the engine. This makes it possible to calibrate it before shipping the ECM to a location where electronic service tools are **not** available. The bench calibration harnesses can also be used for troubleshooting the ECM. Tables 1, 2, and 3 contain items used for the ECM bench calibration procedure.

Some of the bench calibration adapter harnesses include ECM ROM boot capability, (see Table 3). The ROM boot procedure is used when an ECM will **not** accept a calibration or will **not** communicate with the electronic service tool after all other electronic troubleshooting methods have been attempted. The ROM boot procedure resets the software resident in the ECM and is similar to rebooting a personal computer. It is necessary to calibrate the ECM after the ROM boot procedure.

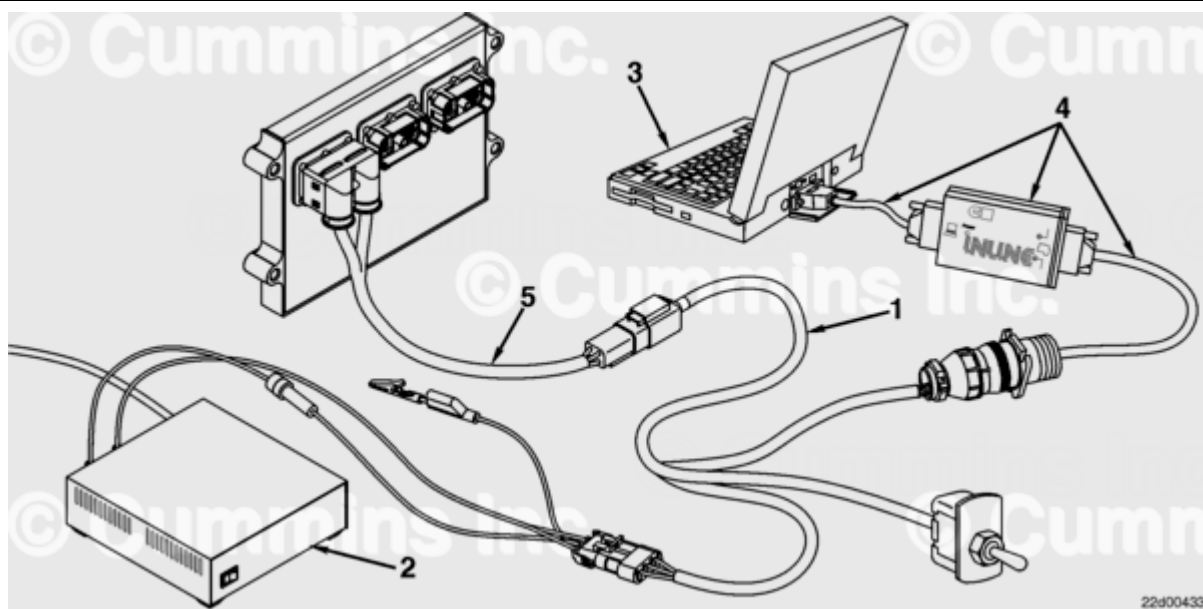


Table 1, ECM Bench Calibration Base Harness			
Item Number	Part Number	Description	Quantity
1	3163151	ECM bench calibration base harness	1
Not Shown	4918881	Base Harness Toggle Switch Repair Kit	1

Table 2, Items used with ECM Bench Calibration Base Harness, Purchased Separately			
Item Number	Part Number	Description	Quantity
2	2892089	Kit, Electrical Power Supply (23 Amp, 120 Volt, 60 Hz with input and output cables)	1
2	2892090	Kit, Electrical Power Supply (23 Amp, 230 Volt, 50Hz with input and output cables)	1
3		INSITE™ electronic service tool-equipped personal computer	1
4	2892092	INLINE™ 6 Adapter Kit	1
4	3163682	INLINE™ II Adapter Kit	1
4	4918416	INLINE™ V Adapter Kit	1
4	5299899	INLINE™ 7 Adapter Kit	1
	2892085	Detachable input cable with BS 1363 plug (For use in the United Kingdom, use only with kit, Part Number 2892090)	1

Note : Previously distributed power supply, Part Number 3164446, may not be able to adequately power the CM2150 and CM2250 modules during calibration. The power supply indicator lamp must remain continuously illuminated during the calibration download. If the indicator lamp extinguishes or flashes, an inadequate power supply is

suspected. The Power Supply Kits listed in the above table will correct these concerns and will power all bench calibration cables distributed by Cummins Inc.

Table 3, Calibration Adapter Harness, Purchased Separately

Item Number	Part Number	ECM	ROM Boot	Engine
5	3162271	CM420	No	B5.9G B5.9LPG C8.3G L10G
5	3164185	CM850	Yes	ISB ISC ISL QSK19 MCRS QSK38 MCRS (see table 4) QSK50 MCRS (see table 4) QSK60 MCRS (see table 4)
5	3164185	CM870	Yes	Signature™ and ISX ISM
5	3164185	CM871	Yes	ISM
5	3164185	CM875	Yes	ISM
5	3164185	CM876	Yes	ISX
5	4918143	CM556B	Yes	B Gas Plus B LPG Plus BGI C Gas Plus L Gas Plus
5	3164867	CM554	Yes	ISC ISL
5	3164868	CM550	Yes	ISB QSB
5	4918145	ECM B	No	CELECT™
5	3164869	ECM C	Yes	CELECT Plus™
5	3164789	CM570	Yes	Signature™ ISX ISM QSM11

Table 3, Calibration Adapter Harness, Purchased Separately

Item Number	Part Number	ECM	ROM Boot	Engine
5	3165030	CM558	Yes	G5.9E GTA8.3SLB G8.3E G855E GTA855E QSK19G KTA19GC KTA19SLB KTA38GC QSK45G QSK60G QSV81G QSV91G
5	3165031	CM700	Yes	QSK19G QSK45G QSK60G QSV81G QSV91G
5	3165062	CM400	No	CENTRY™
5	3165085	CM552	Yes	QST30 (Industrial) 480C-E (Marine)
5	3164046	CM800	No	ISB ^e ISB four-cylinder
5	3163062	CM551D	No	ISB light-duty (Chrysler™)
5	4918142	CM500	Yes	QSK19 QSK23 QSK45 QSK60 QSK78
5	4918583	CM2150	Yes	ISB6.7 QSB3.3 ISC ISL QSK19 MCERS
5	5572672	CM2150	Yes	QSK38 MCERS QSK50 MCERS QSK60 MCERS

Table 3, Calibration Adapter Harness, Purchased Separately

Item Number	Part Number	ECM	ROM Boot	Engine
5	4918802	CM2180/CM2380	Yes	ISL G ISX12 G ISB5.9G ISB6.7G 15N X15N
5	4918938	CM2220	Yes	ISF2.8 ISF3.8 ISB3.9
5	4919009	CM2250	Yes	ISX15 ISX11.9 ISB6.7 QSB6.7 ISC8.3 ISL9 QSL9
5	5298707	CM2330	Yes	K38 K50 K2000 QSK45 QSK60 QSK78 QST30 QSV81 QSV91
5	2892289	CM2350/CM2450	Yes	ISB6.7 ISL9 ISX12 ISX15
5	5298994	CM2350/CM2450	Yes	QSK50 QSK95

Table 3, Calibration Adapter Harness, Purchased Separately

Item Number	Part Number	ECM	ROM Boot	Engine
5	5298534	CM2880	Yes	ISG11 ISG12 ISB/ISD6.7 ISB5.9 ISL8.9 QSB3.9 QSB5.9 QSB6.7 QSC8.3 QSF2.8 QSF3.8 QSL9.3
5	5299150	CM3230	Yes	ISV5.0
5	5394436	CM2358A	Yes	QSK45G QSK60G QSV81G QSV91G
5	5572846	CM2620	Yes	F2.8 F3.8 F4.5 B4.0 D4.0 B4.5 D4.5
5	5572656	CM2670	Yes	B6.2 B6.7 L9 X11 X12 X13 Z14

Table 4, Calibration adapter harness used on engines with multiple ECMs. Purchased separately (items not shown) (To be used with appropriate calibration adapter harness)

Part Number	Description	Quantity
4919064	Bench Calibration Kit*	1
4919022	Multiple module harness	1

*(Bench Calibration Kit, Part Number 4919064, contains one, Part Number 4919022, multiple module harness, also purchased separately, and one, Part Number 4918894, multiple module switch harness, **not** sold separately.)

Related Procedures

05-019-427 (/qs3/pubsys2/xml/en/procedures/05/05-019-427.html)	Engine Control Module ROM Boot (/qs3/pubsys2/xml/en/procedures/05/05-019-427.html)
---	---

Last Modified: 12-May-2023
